

www.efyexpo.com

Dealers & Distributors Solicited

Hundreds of dealership opportunities await you at **Electronics For You Expo 2012.**

Electronics For You expo 2012
Innovate. Design. Manufacture. Source. 16 to 18 February 2012
Pragati Maidan, New Delhi

16 to 18 Feb 2012
Pragati Maidan, New Delhi

Book your tickets and hotels to ensure that you're there for this mega event.

An **EFY GROUP** EVENT

Q. How does voltage regulator LM317 work and how to use it? Also, how to get the output voltage above 40V using regulators?

Anil Kunchal
Kadapa

A. The LM117 series of adjustable three-terminal positive voltage regulators is capable of providing regulated voltage in output range of 1.2V to 37V. The regulator is exceptionally easy to use. It requires only two external resistors to set the output voltage. Further, both line and load regulation are better than standard fixed regulators'.

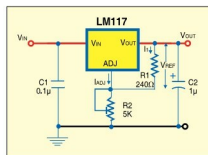


Fig. 1: LM117 in operation

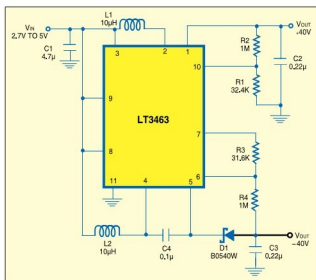


Fig. 2: 2.7V to ±40V dual-output converter

The regulator comes in standard transistor packages, which can be easily mounted and handled. In addition to higher performance than fixed regulators'. LM117 series offers full overload protection. Included on the chip are current limiter, thermal overload protection and safe-area protection.

Normally, no capacitor is needed unless the device is situated more than 15 cm (6 inches) away from the input filter capacitors, in which case an input bypass capacitor (C1) is needed. An optional output capacitor (C2) is added to improve transient response. The adjustment terminal can be bypassed to achieve very high ripple-rejection ratios, which are difficult to achieve with standard three-terminal regulators.

In operation, as shown in Fig. 1, the LM117 develops a nominal 1.25V reference voltage (V_{REF}) between the output and adjustment terminal. The reference voltage is applied across program resistor R1. Since the voltage is constant, a constant current (I_{ADJ}) flows through output set resistor R2, giving an output voltage of:

$$V_{OUT} = 1.25 (1 + R2/R1) + I_{ADJ} R2$$

The LM317 also works as a simple adjustable switching regulator and programmable output regulator. By connecting a fixed resistor between the adjustment pin and output it can also be used as a precision current regulator. For applications requiring greater output current, LM150 series (3A) and LM138 series (5A) are used.

The LM317 is unable to provide 40V DC from a lower voltage on its own. For that, a boosting arrangement in the form of DC-to-DC converter is needed. Fig. 2 shows a circuit based on LT3463 for ±40V from 2.3V-15V input voltage range. For details, refer the datasheet of the device.

Answers compiled by Col (Retd) N.C. Pande.

Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase I, New Delhi 110020 (E-mail: editsec@efymag.com) and should include name and address of the sender.